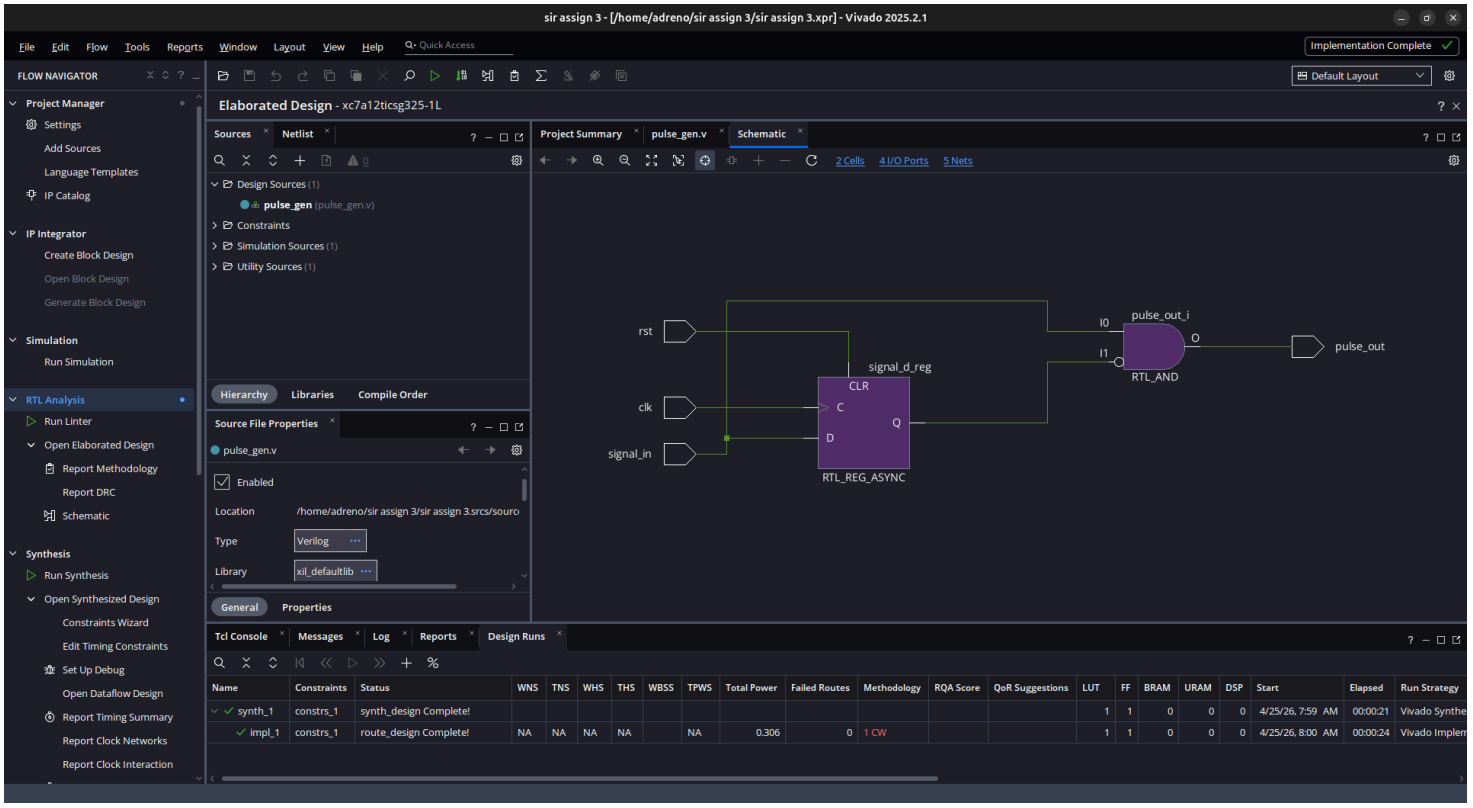
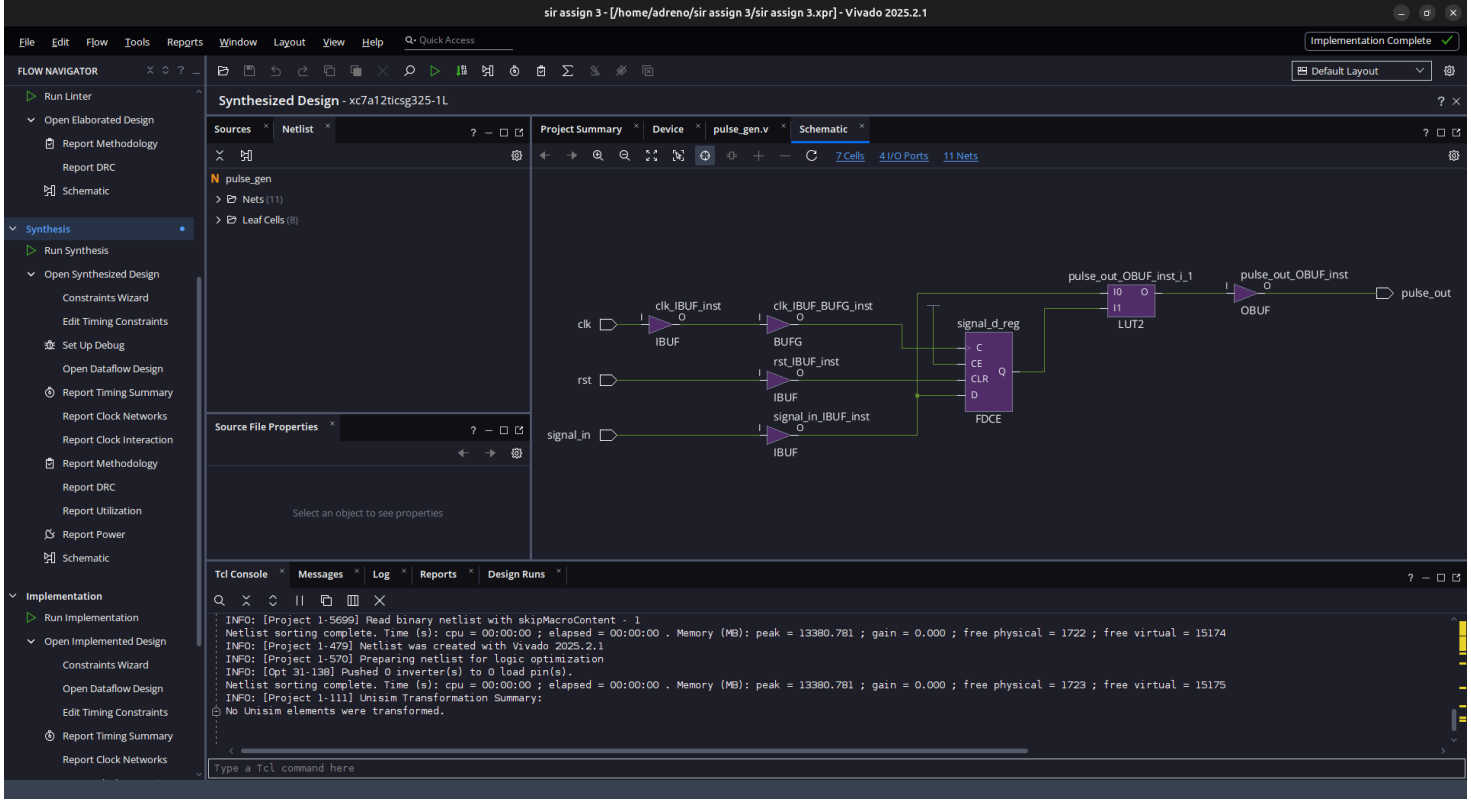
Pulse Generator Single Clock

```
module pulse_gen (
input clk,
input rst,
input signal_in,
output pulse_out
);

reg signal_d;

always @(posedge clk or posedge rst) begin
if (rst)
signal_d <= 0;
else
signal_d <= signal_in;
end

assign pulse_out = signal_in & ~signal_d;

endmodule
```




sir assign 3 - [/home/adreno/sir assign 3/sir assign 3.xpr] - Vivado 2025.2.1
File Edit Flow Tools Reports Window Layout View Help
Quick Access
Implementation Complete
FLOW NAVIGATOR
Run Linter
Open Elaborated Design
Report Methodology
Report DRC
Schematic
Synthesis
Run Synthesis
Open Synthesized Design
Constraints Wizard
Edit Timing Constraints
Set Up Debug
Open Dataflow Design
Report Timing Summary
Report Clock Networks
Report Clock Interaction
Report Methodology
Report DRC
Report Utilization
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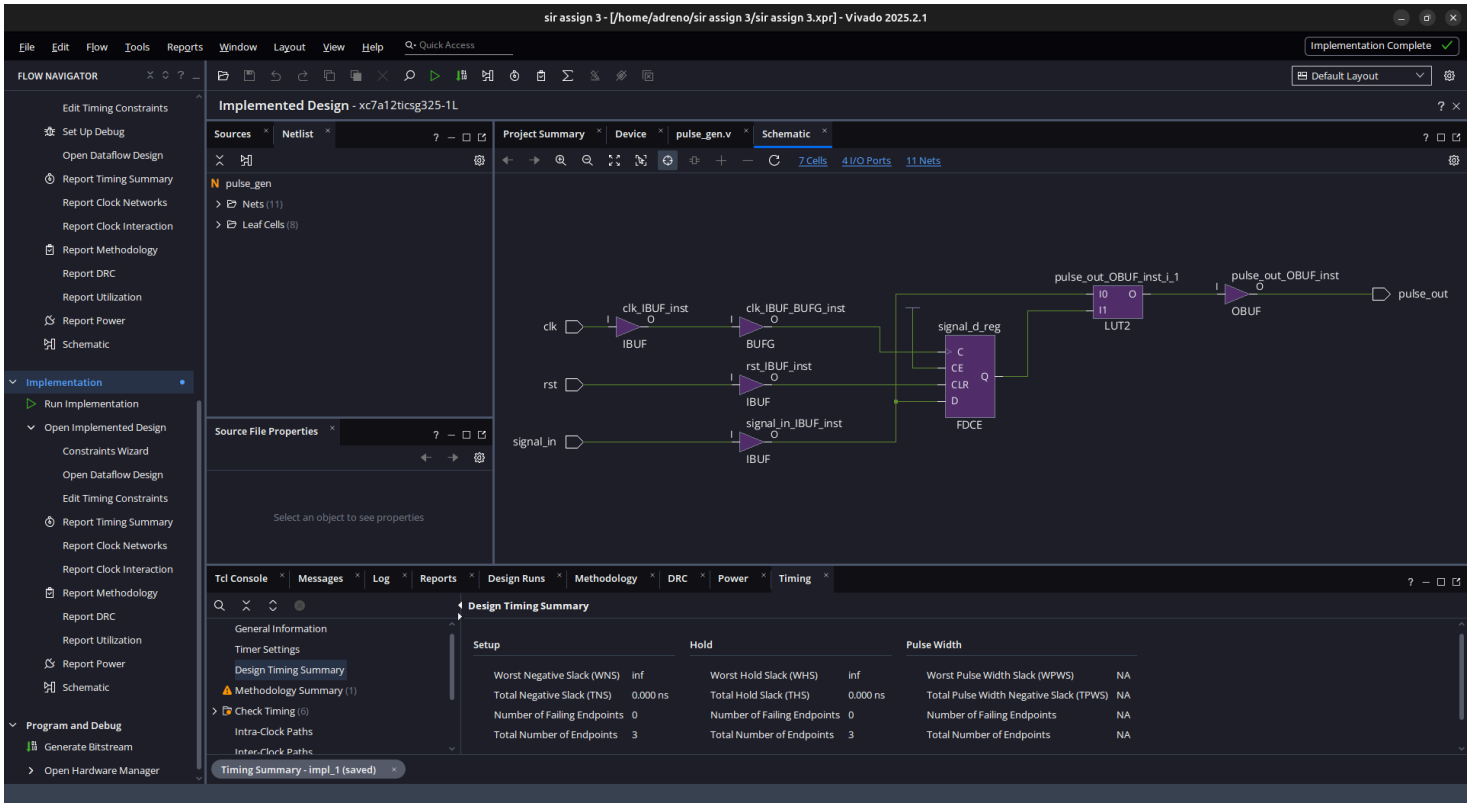
Schematic
Implementation
Run Implementation
Open Implemented Design
Constraints Wizard
Open Dataflow Design
Edit Timing Constraints
Report Timing Summary
Report Clock Networks
Default Layout
Synthesized Design - xc7a12ticsg325-1L
Sources
Netlist
pulse_gen
Nets (11)
Leaf Cells (8)
Source File Properties
Select an object to see properties
Project Summary
Device
pulse_gen.v
Schematic
7 Cells
4 I/O Ports
11 Nets
clk
rst
signal_in
clk_IBUF_inst
IBUF
clk_IBUF_BUFG_inst
BUFG
rst_IBUF_inst
signal_in_IBUF_inst
signal_d_reg
FDCE
C
CE
CLR
D
Q
pulse_out_OBUF_inst_i_1
I0
I1
LUT2
pulse_out_OBUF_inst
OBUF
pulse_out
Tcl Console
Messages
Log
Reports
Design Runs
INFO: [Project 1-5699] Read binary netlist with skipMacroContent - 1
Netlist sorting complete. Time (s): cpu = 00:00:00 ; elapsed = 00:00:00 . Memory (MB): peak = 13380.781 ; gain = 0.000 ; free physical = 1722 ; free virtual = 15174
INFO: [Project 1-479] Netlist was created with Vivado 2025.2.1
INFO: [Project 1-570] Preparing netlist for logic optimization
INFO: [Opt 31-138] Pushed 0 inverter(s) to 0 load pin(s).
Netlist sorting complete. Time (s): cpu = 00:00:00 ; elapsed = 00:00:00 . Memory (MB): peak = 13380.781 ; gain = 0.000 ; free physical = 1723 ; free virtual = 15175
INFO: [Project 1-111] Unisim Transformation Summary:
No Unisim elements were transformed.
Type a Tcl command here


sir assign 3 - [/home/adreno/sir assign 3/sir assign 3.xpr] - Vivado 2025.2.1
File Edit Flow Tools Reports Window Layout View Help
Quick Access
Implementation Complete
FLOW NAVIGATOR
Edit Timing Constraints
Set Up Debug
Open Dataflow Design
Report Timing Summary
Report Clock Networks
Report Clock Interaction
Report Methodology
Report DRC
Report Utilization
Report Power
Schematic
Implementation
Run Implementation
Open Implemented Design
Constraints Wizard
Open Dataflow Design
Edit Timing Constraints
Report Timing Summary
Report Clock Networks
Report Clock Interaction
Report Methodology
Report DRC
Report Utilization
Report Power
Schematic
Program and Debug
Generate Bitstream
Open Hardware Manager
Default Layout
Implemented Design - xc7a12ticsg325-1L
Sources
Netlist
pulse_gen
Nets (11)
Leaf Cells (8)
Source File Properties
Select an object to see properties
Project Summary
Device
pulse_gen.v
Schematic
7 Cells
4 I/O Ports
11 Nets
clk
rst
signal_in
clk_IBUF_inst
IBUF
clk_IBUF_BUFG_inst
BUFG
rst_IBUF_inst
signal_in_IBUF_inst
signal_d_reg
FDCE
C
CE
CLR
D
Q
pulse_out_OBUF_inst_i_1
I0
I1
LUT2
pulse_out_OBUF_inst
OBUF
pulse_out
Tcl Console
Messages
Log
Reports
Design Runs
Methodology
DRC
Power
Timing
Design Timing Summary
General Information
Timer Settings
Design Timing Summary
Methodology Summary (1)
Check Timing (6)
Intra-Clock Paths
Inter-Clock Paths
Setup
Worst Negative Slack (WNS) inf
Total Negative Slack (TNS) 0.000 ns
Number of Failing Endpoints 0
Total Number of Endpoints 3
Hold
Worst Hold Slack (WHS) inf
Total Hold Slack (THS) 0.000 ns
Number of Failing Endpoints 0
Total Number of Endpoints 3
Pulse Width
Worst Pulse Width Slack (WPWS) NA
Total Pulse Width Negative Slack (TPWS) NA
Number of Failing Endpoints NA
Total Number of Endpoints NA
Timing Summary - impl_1 (saved)